Chinnolla Koteshwar



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Marital status	Single	**Portfolio**	koteshwarchinnolla.github.io

Research Profile

I am currently pursuing a Master's in Autonomy Technologies at FAU Erlangen-Nürnberg, specializing in robotics frameworks, deep learning, computer vision, and reinforcement learning. Previously, as an Associate Software Engineer, I developed and deployed scalable AI applications—including recommendation systems and agentic workflows—into production environments. This experience solidified my expertise in cloud infrastructure and DevOps, utilizing Kubernetes, Terraform, and robust CI/CD pipelines, alongside gRPC for fast service-to-service communication and GraphQL for efficient data retrieval. During my Bachelor's in Electronics and Communication Engineering, I established a strong foundation in hardware and applied machine learning. This culminated in my final-year thesis, where my team and I developed a natural language-based autonomous navigation vehicle, successfully published at IEEE ICOCT 2025.

Research Interests

Autonomous Robotics	**Natural Language Systems**	**Reinforcement Learning**
Intelligent Agents	**Computer Vision**	**Autonomous Systems**

Relevant Coursework

AI & Machine Learning	Artificial Neural Networks, Machine Learning, Python
Computer Science	Data Structures, Object-Oriented Programming (Java), Cloud Computing, DBMS
IoT & Embedded Systems	Internet of Things, Microcontrollers
Mathematics	Linear Algebra, Differential Calculus, Probability & Stochastic Processes
Control & Signals	Linear Control Systems, Digital Signal Processing, Signals & Systems
Core Electronics	Analog & Digital Circuits, VLSI, Networks

Education

M.Sc. – Autonomy Technologies — **Oct 2026 – Present**
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany

B.Tech – Electronics & Communication Engineering *(Minor: AI/ML)* — **Nov 2021 – May 2025**
Gokaraju Rangaraju Institute of Engineering and Technology (GRIET), Hyderabad
CGPA: **8.39 / 10**

Senior Secondary (Intermediate) — **July 2019 – April 2021**
NSR Impulse Junior College
988 / 1000

Secondary School Certificate — **June 2018 – March 2019**
Matrusri High School
9.8 / 10

Publications & Research

[1] **C. Koteshwar**, M. D. Chaithanya, M. Abdul Kaleem, and A. Vishnubhatla, *"Natural Language Based Autonomous Navigation Vehicle,"* in *Proc. 2025 International Conference on Computing Technologies (ICOCT)*, Bengaluru, India, 2025, pp. 1–7.
DOI: 10.1109/ICOCT64433.2025.11118685 • **Venue:** IEEE ICOCT 2025 • **Keywords:** *LLMs, Computer Vision, Natural Language, Autonomous Navigation*

Work Experience

Associate Software Engineer – AI & DevOps *Anasol Consultancy Services, Hyderabad* Apr 2025 – May 2026

- Built and maintained AI-enabled backend systems for enterprise applications.
- Worked on recommendation systems and agent-based workflows for internal automation.
- Developed scalable infrastructure using AWS, Kubernetes, Terraform, Helm, Redis, Kafka, and microservice design patterns.
- Created reusable open-source Terraform modules and Helm charts for production-ready cloud deployments.

Projects

ApiSynIQ Go, Python, Java, gRPC, PostgreSQL (pgvector), LangChain, LangGraph

- Developed a multi-service framework for natural language interaction with REST APIs, that lets users interact with APIs using natural voice or text instead of traditional forms.
- Designed a multi-service architecture to understand user intent, identify the right endpoint, and generate valid API requests automatically.
- Implemented a semantic API discovery engine with pgvector and RAG, enabling the system to match human language to the correct backend service.
- Created a developer-friendly annotation SDK that turns existing APIs into conversational interfaces with minimal code changes.

Bachelor's Thesis: Natural Language Based Autonomous Navigation Vehicle Python, Raspberry Pi, Arduino, OpenCV, A*, Docker, LLMs

- Developed an autonomous vehicle that interprets spoken or text commands into task-specific navigation workflows in indoor environments.
- Integrated LLM-based planning and response modules to interpret user intent, ask clarification questions, and manage multi-step tasks dynamically.
- Implemented A* path planning with a custom motion decoder, enabling accurate obstacle-aware navigation and object delivery between locations.
- Published the work at IEEE ICOCT 2025 for its approach to combining conversational AI with robotic task execution.

Cloud Infrastructure Automation Toolkit Terraform, Helm, Kubernetes, AWS, ArgoCD, Istio - Terraform - HELM

- Built reusable Terraform modules and Helm charts to automate cloud infrastructure provisioning and Kubernetes deployments across AWS environments.
- Designed production-ready templates for EKS, ECS, VPC, RDS, ALB, Kafka, Redis, and microservices, reducing repetitive setup for scalable systems.
- Integrated GitOps workflows with ArgoCD and Helm for consistent, version-controlled deployments from infrastructure to application layer.

NLP Architecture Evolution Study Python, TensorFlow, PyTorch, Scikit-learn, Transformers - BERT - NLP Techniques - RNNs

- Developed three connected NLP projects covering spam classification, BERT pre-training, and sentiment analysis to understand the evolution from statistical methods to deep contextual learning.
- Implemented and compared Bag of Words, TF-IDF, Word2Vec, Bi-LSTM, and BERT, observing how each architecture improved language understanding and prediction accuracy.
- Built models from the ground up to study internal behavior of each technique rather than treating them as black-box libraries.

OpenCV Computer Vision Learning Repository Python, OpenCV, YOLOv8, NumPy

- Built a practical computer vision repository exploring core image processing concepts and real-time object detection workflows.
- Developed a vehicle counting system using line-cross detection and object tracking to monitor cars in video streams.
- Integrated YOLOv8-based object detection on images and videos to study modern deep learning approaches in computer vision.

Technical Skills

Programming	Python, C++, Java, Go
AI & Research	Deep Learning, TensorFlow, PyTorch, OpenCV, keras, NLP, Reinforcement Learning, RAG, LangChain, LangGraph
Robotics	ROS, Raspberry Pi, Arduino, Computer Vision
Cloud & Infrastructure	AWS, Kubernetes, Terraform, Helm, Docker, Kafka, Redis

Leadership & Activities

- **Mentorship (Advanced Academic Center):** Mentored 4 students in AI/ML, guiding them from initial concepts to successful project implementations and final submissions.
- **Problem Solving:** Solved **480+** problems across Java, Python, C++, and Go on LeetCode.

Hackathons

- Participated in 4 technical hackathons.
- **Winner:** AAC Product Demand Prediction Challenge.
- **CBIT SUDHEE 23:** ML-integrated healthcare system.
- **EPITOME 23:** MedTech.
- **CHALLENGE ACI:** Blockchain-based Electronic Health Record Management.

Training

- Agentic AI – Krish Naik Academy.
- AWS Cloud – AWS Skill Builder.
- Machine Learning – Udemy.

Language Proficiency & Test Scores

English **2026**

IELTS 7.0 (CEFR Level C1)

GRE 322 (V: 163, Q: 159, AW: 3.0)

Telugu

Native N/A

Hindi

Fluent N/A